

5 (**long**) Figure 5 shows a cantilevered structure anchored to the ground at frictionless pin-joints A and B. A vertical load of 10 kN is applied downwards at joint D. The figure is drawn to scale and the bar lengths shown in the figure are in metres. All bars have the same cross-sectional area ( $A$ ) and Young's modulus ( $E$ ) so that  $AE = 10^7$  N.

- (a) Find the forces in all the bars of the structure. [10]
- (b) Find the vertical displacement of joint E. [20]

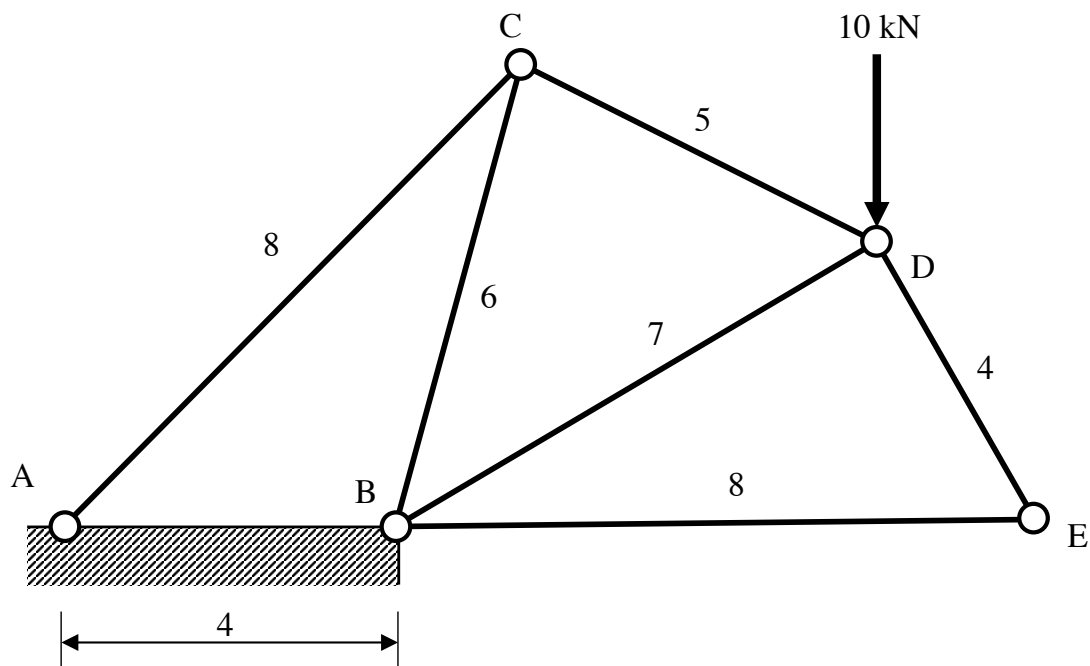


Fig. 5